

Week 1

0. Introduction and Orientation

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COMP.2800 – Software Development [2026W]

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University of Windsor

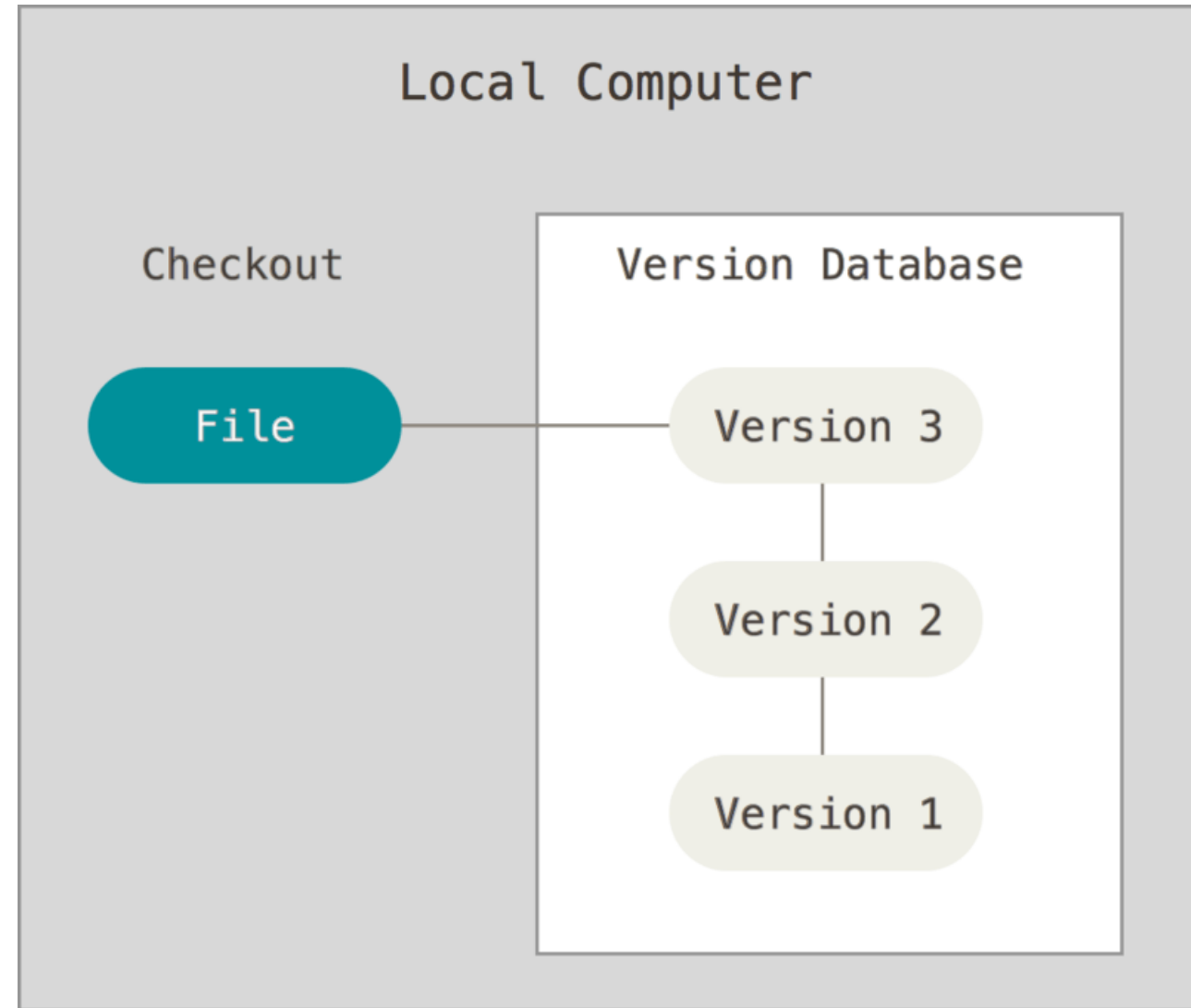


What is Git?



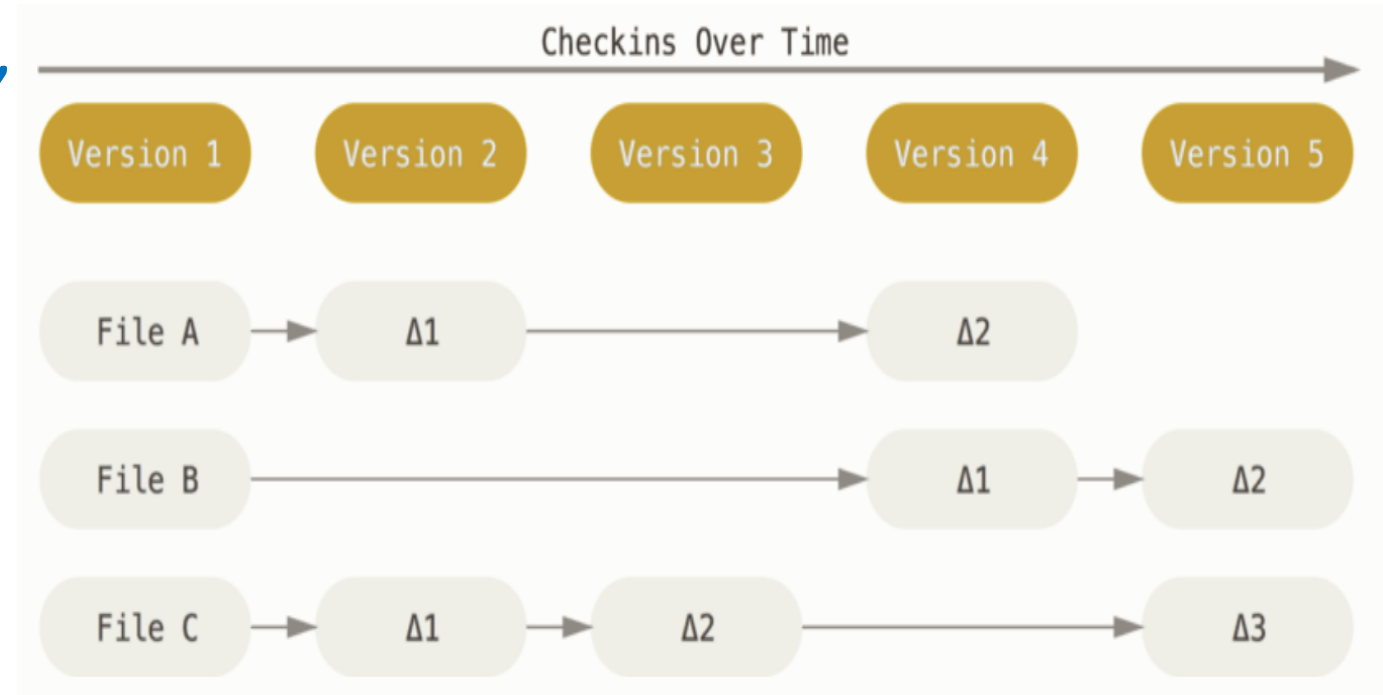
What is GIT?

- **Git is a distributed version control system.**
- But what is a version control system (VCS)?



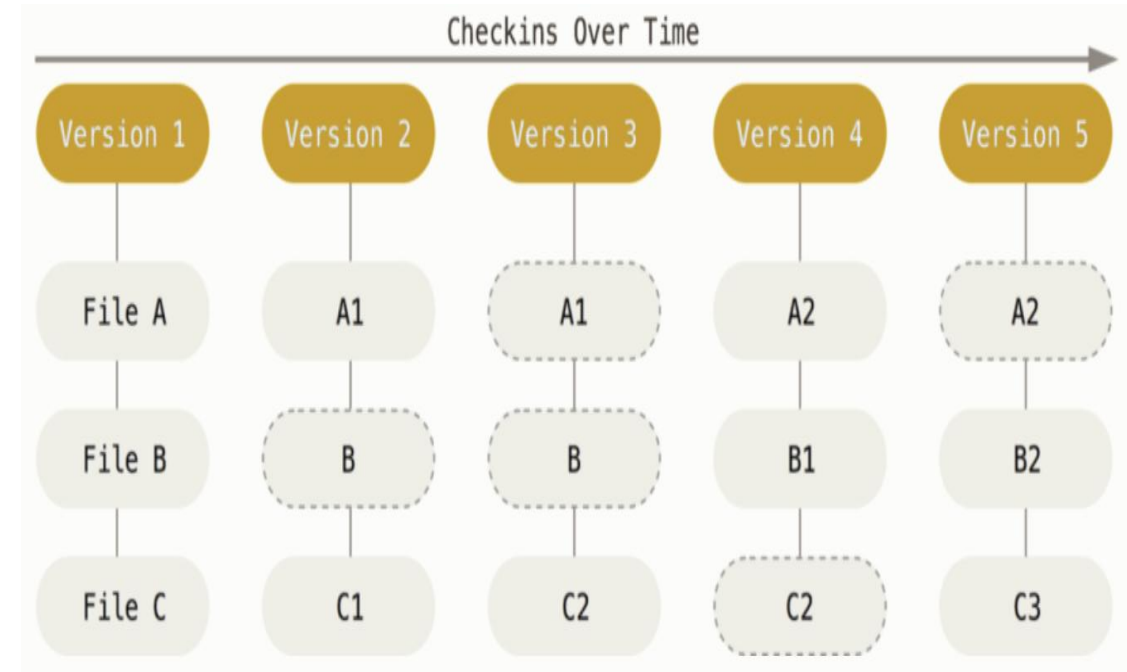
Delta based VCS

- Stores changes as deltas.
- Efficiently reconstructs file versions.
- Examples: CVS, Subversion, and Perforce.



Snapshot Based VCS

- Tracks complete project state including all files and directories
- Snapshots allow for efficient storage and retrieval
- Example GIT





Why do we need a VCS?



Why do we need a VCS?

- Collaboration and Teamwork
- History Tracking and Auditing
- Code Stability and Release Management
- Conflict Resolution
- Backup and Disaster Recovery



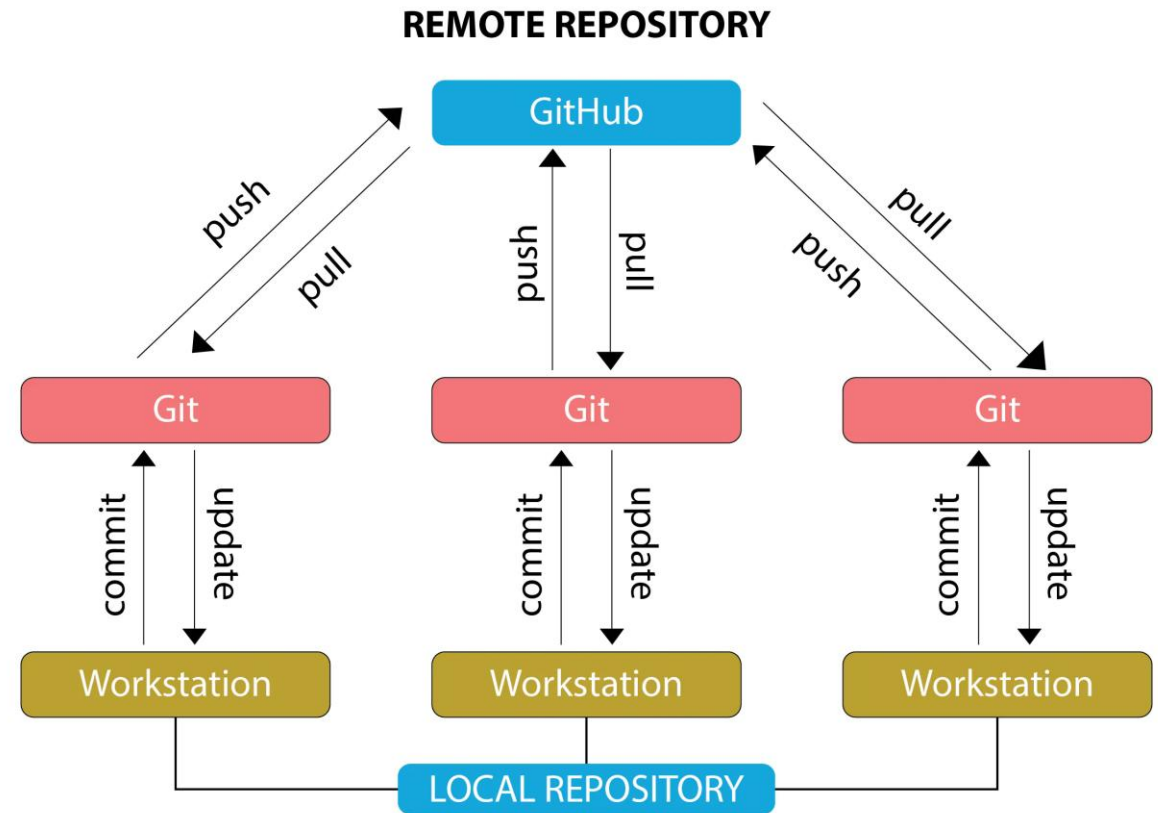


About Git



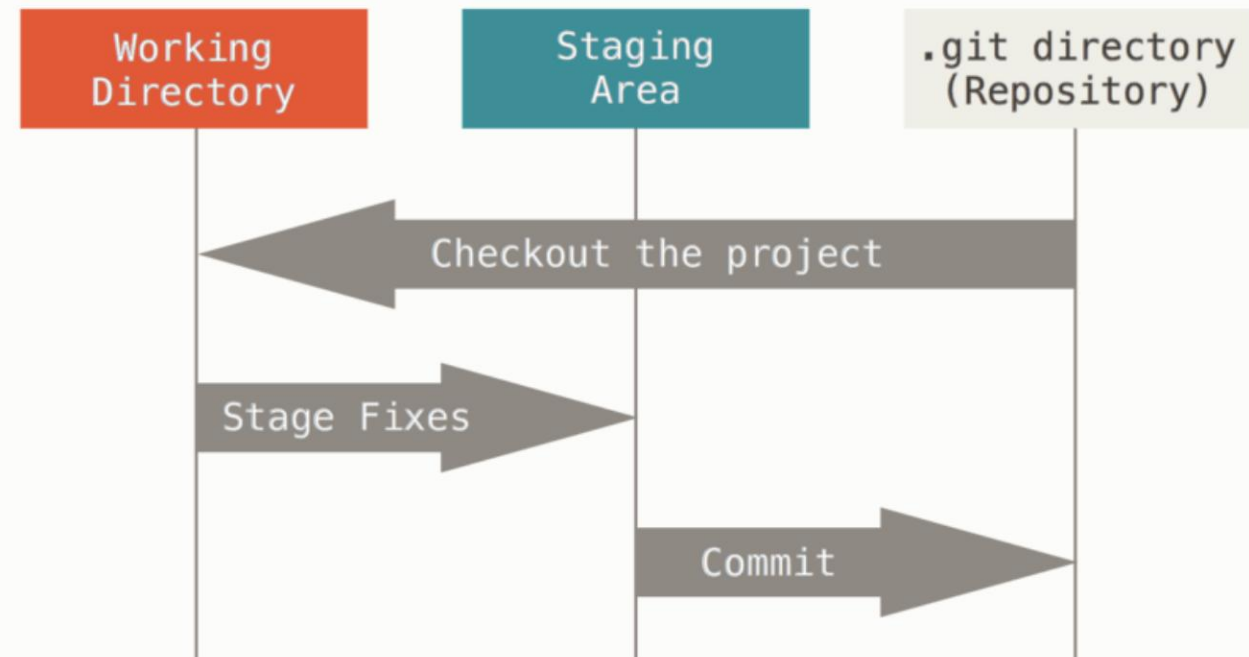
More on GIT

- A snapshot-based VCS
- Has a central remote repository view changes
- GitHub, GitLab, etc. are example of central remote repositories



File States in GIT

- **Modified:** Files are changed but are not committed yet
- **Staged:** Modified files are marked to go into next commit snapshot
- **Committed:** Modified files are stored in a local database





Workflow in Git



Basic workflow in Git

- Create a new repository or clone a repository
- Modify files in the working tree
- Added the modified files to the staged area
- Commit the changes that permanently store the files in the staged area to the Git directory





Installing Software



● Installing GIT

○ Installing on Linux

```
$ sudo apt update  
$ sudo apt install git
```

Installing on Mac

```
https://sourceforge.net/projects/git-osx-installer/files/
```

Installing on Windows

```
https://gitforwindows.org/
```

Setting up GIT

Check version

```
$ git --version  
git version 1.7.1
```

Command help

```
$ git help <verb>  
$ git <verb> --help  
$ man git-<verb>
```

Example

```
$ git log --help  
$ git help log
```

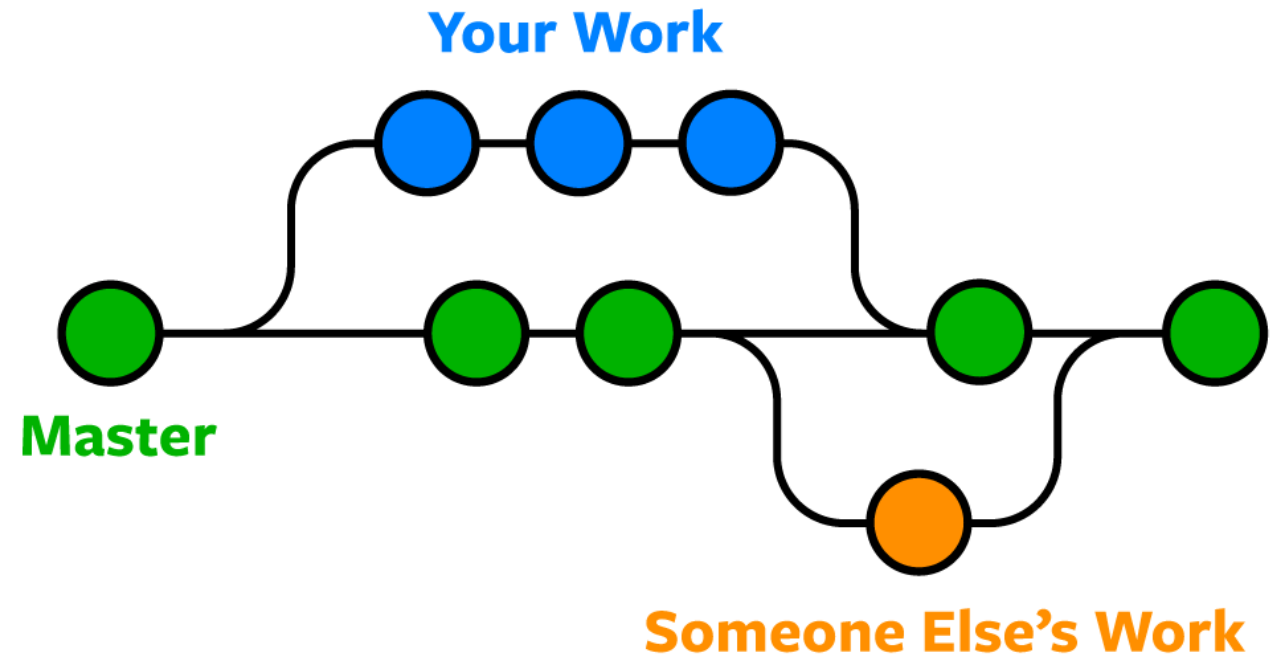
Installing VS-Code

- <https://code.visualstudio.com/docs/setup/setup-overview>
- Follow the steps to install VS Code in your pc/laptop



Git Branching

- A git branch is a pointer to a tree, and it is movable.
- The default branch name in Git is **main**.



Demo on Basic Workflow of GIT



Basic workflow of git using VS Code





Commits



Git commit history

```
$ git log
commit ca82a6dfff817ec66f44342007202690a93763949
Author: Scott Chacon <schacon@gee-mail.com>
Date:   Mon Mar 17 21:52:11 2008 -0700
```

Change version number

```
commit 085bb3bcb608ele8451d4b2432f8ecbe6306e7e7
Author: Scott Chacon <schacon@gee-mail.com>
Date:   Sat Mar 15 16:40:33 2008 -0700
```

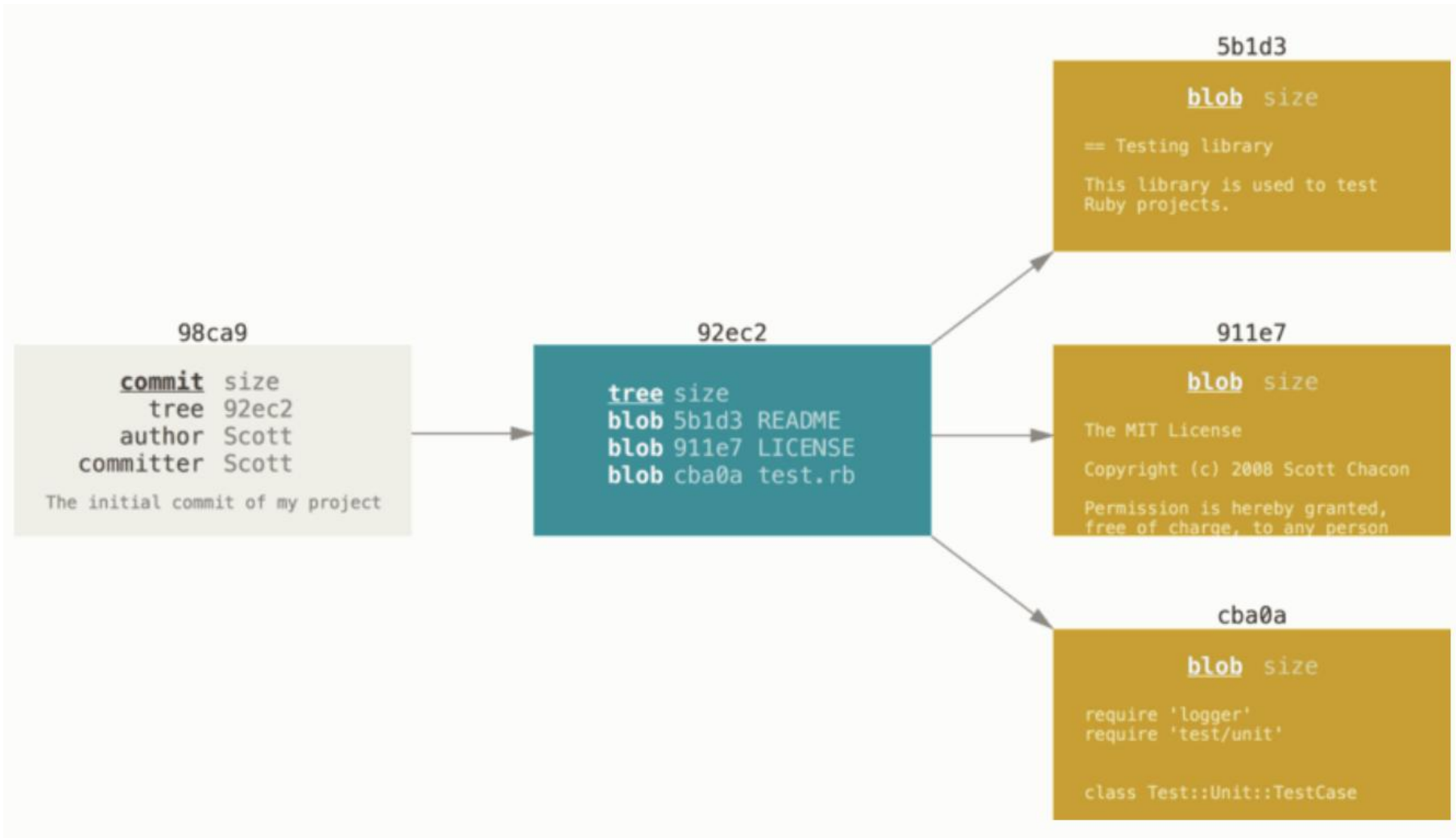
Remove unnecessary test

```
commit a11bef06a3f659402fe7563abf99ad00de2209e6
Author: Scott Chacon <schacon@gee-mail.com>
Date:   Sat Mar 15 10:31:28 2008 -0700
```

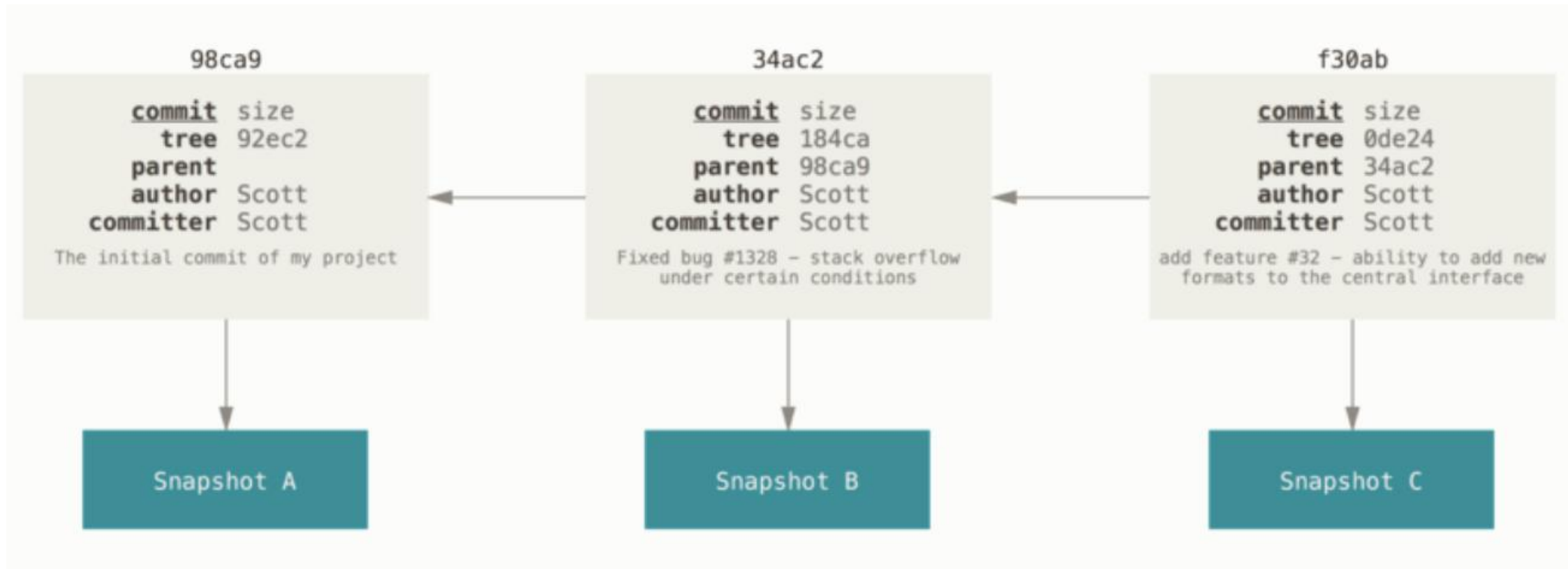
Initial commit



Git Commit tree



Git commit trees

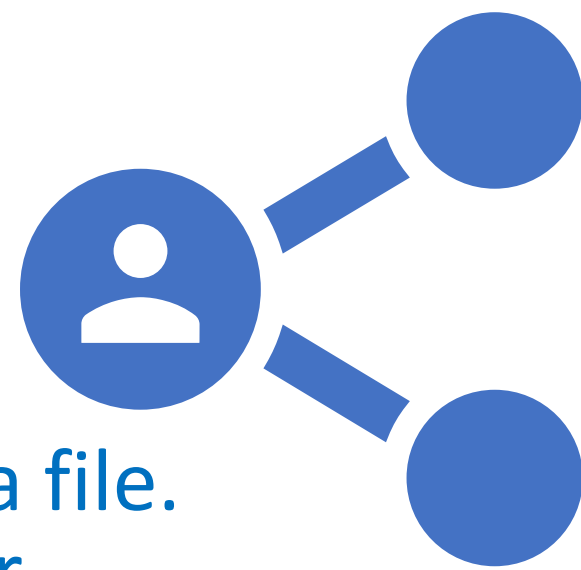




Merging



Merge Conflict



- Conflicts arise when:
 - Two developers change the same lines in a file.
 - One developer deletes a file while another modifies it.
- Git cannot automatically determine the correct version in these cases.
- Conflicts only affect the developer conducting the merge; the rest of the team remains unaware.
- Git marks the file as conflicted and halts the merging process

Merge Conflict Using VS Code



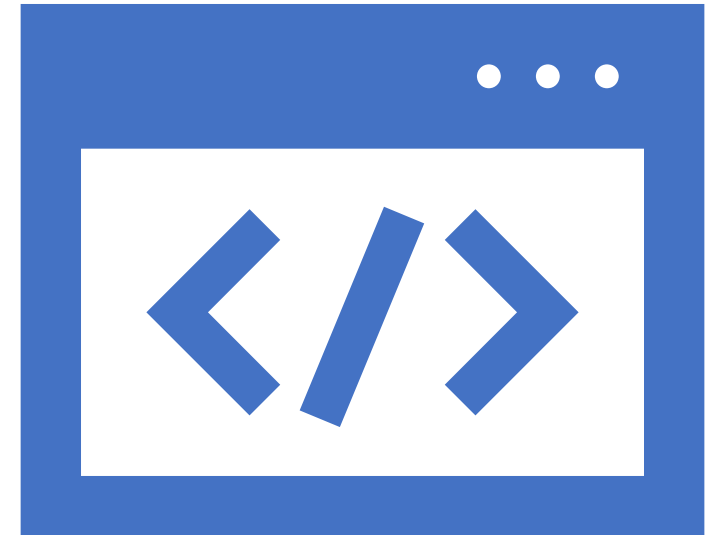


References



References

- [Using Git with Visual Studio Code \(Official Beginner Tutorial\) \(youtube.com\)](#)
- [Git \(git-scm.com\)](#)



Thank you!
Questions?

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